

Mind at Work: Detecting Agency through Assembly and Action

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Abstract

This monograph presents an extended framework for detecting mind-like agency by integrating the Effective Assembly Index for Mind Recognition (EAIMR) with the praxicon concept from cognitive science. EAIMR quantifies the improbability of a system’s structure arising from unguided processes, adjusted for efficiency and amplified by coherence, while the praxicon provides a structured repository of action representations bridging perceptual, semantic, and motor domains. By combining these, we propose a robust method to identify agentic intelligence in neural systems, collective organisms, ecosystems, and artificial intelligence, avoiding anthropocentric biases. The framework is formalized using information theory, graph theory, category theory, and sheaf theory, with applications in neuroscience, AI alignment, ecological cognition, and cross-cultural studies. We contrast this approach with competing theories, provide detailed case studies, and discuss ethical implications for recognizing non-human minds. This extended version includes prerequisite explanations for foundational concepts and advanced interpretations using category and sheaf theory.

1 Introduction

The detection of agency and mind-like processes in complex systems represents a fundamental inquiry spanning cognitive science, philosophy of mind, artificial intelligence, and ecology. Agency, in this context, refers to the capacity of a system to pursue goals through coordinated actions that deviate from equilibrium states predicted by unguided physical laws. The Effective Assembly Index for Mind Recognition (EAIMR) serves as a quantitative tool to measure the improbability of such coordinated structures, while the praxicon offers a cognitive architecture for representing actions in a manner that bridges embodied and symbolic domains. This monograph integrates these elements to form a unified framework capable of identifying agency across diverse substrates, from biological neural networks to collective ecological systems and artificial agents.

To appreciate this integration, it is essential to understand prerequisite concepts. Information theory, pioneered by Claude Shannon, quantifies uncertainty and structure in systems through measures like entropy, which represents the average information content or disorder in a probability distribution [?]. Complexity measures, such as Kolmogorov complexity, define the shortest program capable of generating a given structure, providing a basis for assessing improbability [?]. Assembly theory extends these ideas to biological and technological contexts by counting the minimal steps required to construct complex objects from simple building blocks [?]. The praxicon, rooted in neuropsychology, models how the brain stores and retrieves action knowledge, distinguishing between perceptual recognition and motor execution [?]. By weaving these threads together, our framework addresses the challenge of detecting agency without relying on anthropocentric markers like language or self-report.

2 Historical and Conceptual Foundations

2.1 Prerequisites: Complexity and Information in Natural Systems

Before delving into assembly theory, consider the foundational role of complexity in distinguishing living and agentic systems from inert matter. Murray Gell-Mann’s work on effective complexity highlights how systems exhibit regularities that are neither too random nor too ordered, a balance often indicative of adaptive processes [?]. Robert Hazen’s research on mineral evolution and biocomplexity introduces functional

information as a metric for the emergence of life-like properties, where improbable configurations serve functional roles [?]. These ideas set the stage for assembly theory, which quantifies the "assembly index" as the minimal number of operations needed to produce a structure, emphasizing reuse and hierarchy in evolved systems.

2.2 Assembly Theory and Complexity

Assembly theory, developed by Sara Imari Walker, Leroy Cronin, and collaborators, provides a formal measure of complexity by calculating the shortest path to assemble a molecule or structure from basic units [??]. It has applications in astrobiology and origins-of-life research, where high assembly indices correlate with biotic processes. EAIMR builds upon this by incorporating adjustments for real-world efficiencies and coherence, enabling its use as a detector for mind-like agency [?].

2.3 Praxicon: Action Representation in Cognition

The praxicon emerges from studies of apraxia, a disorder impairing purposeful movements despite intact motor function. Prerequisite knowledge includes the distinction between ideational apraxia (conceptual deficits) and ideomotor apraxia (execution deficits), as modeled by Rothi and Heilman [?]. The praxicon is conceptualized as a dual-component system: the input praxicon for recognizing actions via visual-gestural cues, and the output praxicon for producing motor engrams [?]. Extensions to AI involve grounding symbolic language in sensorimotor primitives, addressing the symbol grounding problem [??].

2.4 Competing Approaches

To contextualize our framework, compare it with alternatives. Integrated Information Theory (IIT) posits consciousness as integrated information Φ , but it requires predefined system boundaries and struggles with distributed agency [?]. The Free Energy Principle (FEP) frames agents as minimizers of variational free energy, modeling predictive processing but lacking direct measures of structural improbability [?]. Enactive cognition emphasizes autopoiesis and sensorimotor contingencies, yet it often remains qualitative [?]. Our EAIMR-praxicon approach complements these by providing computable metrics grounded in assembly and action representation.

3 Formal Definitions

3.1 Prerequisites: Probabilistic Models and Generative Distributions

Understanding EAIMR requires familiarity with generative models. A null model $P_0(X)$ represents mindless processes, such as stochastic diffusion or self-organization without goals, while a mind-like model $P_1(X)$ incorporates agency, such as optimization toward functional setpoints. These distributions allow computation of likelihoods and surprises, foundational to information-theoretic analyses [?].

3.2 EAIMR Definition

For a system X described over an alphabet Σ , EAIMR is:

$$\text{EAIMR}(X) = \frac{A(X)}{R(X) \cdot P(X) \cdot D(X)} \cdot C(X),$$

where: - $A(X)$: Raw Assembly Index, the minimal steps in an assembly grammar [?]. - $R(X)$: Reuse factor, quantifying modular repetition. - $P(X)$: Parallelism factor, accounting for concurrent processes. - $D(X)$: Degeneracy factor, reflecting alternative configurations achieving similar functions. - $C(X)$: Coherence factor, amplifying goal-directed integration.

High values suggest agency, as the structure is improbable under $P_0(X)$ but plausible under $P_1(X)$.

3.3 Praxicon Definition

The praxicon is formalized as a bipartite graph $G_{\mathcal{P}} = (V_A, V_S, E)$, with action vertices V_A (e.g., motor programs), semantic vertices V_S (e.g., conceptual labels), and edges E encoding mappings [?]. This structure facilitates bidirectional translation between perception and action, essential for embodied cognition [?].

4 Mathematical Integration

4.1 Information-Theoretic Foundations

Self-information under the null: $I_0(X) = -\log P_0(X)$. The likelihood ratio $\Lambda(X) = P_1(X)/P_0(X)$ quantifies preference for agentic models. For praxicon integration, action entropy reduction is:

$$\Delta H_{\mathcal{P}} = H(\mathcal{P}_0) - H_{\text{obs}}(\mathcal{P}),$$

updating coherence as $C(X) \approx 1 + \alpha \Delta H_{\mathcal{P}} / H(\mathcal{P}_0)$ [??]. Kolmogorov complexity approximates $A(X)$, with praxicon engrams compressing descriptions [?].

4.2 Graph-Theoretic Formalism

The praxicon graph $G_{\mathcal{P}}$ allows coherence via spectral properties, e.g., eigenvalue stability for robustness. EAIMR identifies low-entropy subgraphs indicative of agency [?].

4.3 Category-Theoretic Interpretation

Category theory provides a formal language for describing structures and transformations in abstract terms, making it particularly suitable for modeling cognitive processes that involve mappings between different domains [??]. A category consists of objects (e.g., perceptual states, motor actions) and morphisms (transformations or relations between objects) that satisfy associativity and identity axioms. Functors map between categories while preserving structure, and natural transformations provide morphisms between functors.

In our framework, the praxicon can be viewed as a functor $F : \mathcal{C}_{\text{perc}} \rightarrow \mathcal{C}_{\text{motor}}$, where $\mathcal{C}_{\text{perc}}$ is the category of perceptual inputs (objects as sensory data, morphisms as perceptual transformations) and $\mathcal{C}_{\text{motor}}$ is the category of motor outputs (objects as action states, morphisms as motor adjustments), mediated through a semantic category $\mathcal{C}_{\text{sem}}$. This functorial representation captures how perceptual inputs are systematically transformed into motor actions via semantic grounding.

EAIMR, in this context, quantifies the "improbability" or cost of natural transformations $\eta : F \Rightarrow G$, where G represents alternative mappings (e.g., under null dynamics). High EAIMR corresponds to scenarios where the observed natural transformation (the agentic mapping) is highly specific and unlikely under random perturbations, reflecting the coordinated agency. For instance, in neural systems, category-theoretic models can describe how cognitive development involves composing functors for increasingly complex action representations [?]. This interpretation aligns with structuralist views in cognitive science, emphasizing relational structures over material substrates [?].

4.4 Sheaf-Theoretic Interpretation

Sheaf theory extends category theory by addressing how local data on a topological space can be consistently glued into global structures, offering a powerful tool for modeling distributed systems where agency emerges from multi-scale interactions [?]. A sheaf $\mathcal{F}$ on a space X assigns to each open set $U \subseteq X$ a set $\mathcal{F}(U)$ (local data) with restriction maps that satisfy gluing conditions: local sections that agree on overlaps can be uniquely combined into a global section.

In biological and ecological contexts, sheaf theory models uncertainty and resilience in complex networks [?]. For our framework, define a sheaf $\mathcal{F}$ on the topological space representing the system's components (e.g., neurons in a circuit or species in an ecosystem), where sections over open sets encode local praxicon fragments—action representations valid in subsystems. The gluing axiom ensures that local actions cohere into global behaviors, such as coordinated foraging in ant colonies or nutrient cycling in forests.

EAIMR quantifies obstructions to trivial gluing, often captured by sheaf cohomology groups $H^k(X, \mathcal{F})$. Non-zero cohomology indicates "twists" or global inconsistencies under null models, corresponding to high improbability and thus agency. For example, in ecology, forest nutrient networks form a sheaf where local flows (sections) glue into global cycles; high H^1 (non-exact cycles) elevates EAIMR, detecting emergent intelligence [?]. In biology, bioelectric signaling can be sheafified, with cohomology measuring scale-free cognition [??]. This approach bridges local computations to global agency, providing a mathematical basis for resilience under perturbations.

5 Worked Examples

5.1 Neural Tissue

In neural systems, high $A(X)$ arises from precise synaptic architectures, with praxicon engrams in cortical columns reducing entropy. Simulations of mirror neurons show effector-specific mappings [?]. Step-by-step: Compute $A(X)$ via connection counts, adjust discounts, amplify with coherence from goal pursuit. Entropy curve: Sharp reduction in action space.

5.2 Termite Mound

Moderate assembly in tunnels, high coherence in thermoregulation. Collective praxicon via pheromone-guided motifs [?]. Graph: Nodes for worker actions linked to colony semantics. Data from self-organization models [?].

5.3 Forest Ecosystem

Extensive macro-assembly in symbiotic networks, praxicon-like routing. High degeneracy in species roles, coherence over timescales [?]. Lattice diagram: Interconnected trophic levels.

5.4 Negative Case: Crystal Growth

Regular lattice growth yields low $C(X)$, high predictability under P_0 , resulting in low EAIMR [?].

6 Applications

6.1 Neuroscience

Applies to apraxia diagnosis and action simulation, leveraging praxicon deficits [??].

6.2 AI and Robotics

Praxicon dictionaries ground LLMs in robotics; EAIMR detects emergent agency [??].

6.3 Ecological and Collective Minds

Identifies intelligence in mounds, colonies, forests via distributed praxicons [??].

6.4 Cross-Cultural Cognition

Rituals encode cultural praxicons, detectable as high-EAIMR action repositories [?].

7 Limits and Future Research

Challenges include approximating uncomputable complexities and validating sheaves in empirical data. Future directions: Integrate with bioelectric models [?] and test in multi-agent AI.

8 Ethical and Philosophical Implications

Promotes recognition of non-human minds, informing AI ethics and ecology [?]. Aligns with circumstantial selfhood [?] and processual becoming [??].

9 Decision Rule

Systems with $\text{EAIMR}(X) \geq \tau$ (calibrated on mindless references) are classified agentic, using proxies like graph entropy.

10 Conclusion

This framework advances agency detection through assembly, action, and advanced mathematical lenses, fostering interdisciplinary insights.

A Mathematical Appendix: EAIMR, Information, and Improbable Order

A.1 Preliminaries & Notation

Let X denote a structure/system (e.g., termite mound, forest network, neural microcircuit) described over an alphabet Σ .

Let P_0 be a null (mindless) generative model for X (e.g., stochastic physics + simple self-organization) with distribution $P_0(X)$.

Let P_1 be a mind-like generative model for X (agentic/goal-directed) with distribution $P_1(X)$.

Raw Assembly Index $A(X)$: minimum step count in a prescribed assembly grammar to build X from primitives.

EAIMR (from your definition):

$$\text{EAIMR}(X) = \frac{A(X)}{R(X) P(X) D(X)} C(X)$$

A.2 Connection to Kolmogorov Complexity

Let $K(X)$ be prefix Kolmogorov complexity (shortest universal description of X).

In general, lower $K(X)$ higher compressibility (simple regularities).

Many mind-built artifacts have mixed profiles: globally compressible high-level plan + low-level details that are costly to specify.

A useful proxy is two-part MDL (minimum description length):

$$L(X) \approx L(\text{model}) + L(\text{residuals} \mid \text{model})$$

Heuristic link (non-rigorous but actionable):

$$\text{EAIMR}(X) \propto \frac{K_{\text{plan}}(X)}{R P D} \cdot C \quad \text{with} \quad K_{\text{plan}}(X) \lesssim K(X)$$

A.3 Statistical Surprise & Likelihood Ratios

Define the self-information under the null:

$$I_0(X) \equiv -\log P_0(X)$$

$$\Lambda(X) \equiv \frac{P_1(X)}{P_0(X)} = \exp \left(\log P_1(X) - \log P_0(X) \right).$$

Key idea: High EAIMR should correlate with large $\Lambda(X)$ and a high $I_0(X)$ (i.e., X is very unlikely under mindless dynamics but plausible under mind-like dynamics).

A practical surrogate score you can compute or estimate:

$$S(X) \equiv I_0(X) + \lambda \phi(X)$$

$$\text{EAIMR}(X) = g(S(X)), \quad g \uparrow$$

A.4 Entropy Reduction, Work, and Directed Assembly

Let $\mathcal{M}$ be macrostate features (e.g., ventilation topology, nutrient routing, neural coding). Define macro-entropy under P_0 and observed for X .

Directed assembly manifests as entropy reduction:

$$\Delta H \equiv H(\mathcal{M}) - H_{\text{obs}}(\mathcal{M}) > 0$$

A convenient coherence multiplier:

$$C(X) \approx 1 + \alpha \frac{\Delta H}{H(\mathcal{M})}$$

A.5 Reuse, Parallelism, Degeneracy as Information Discounts

Treat these as code-length discounts relative to the raw assembly plan.

Reuse : library motifs shorten description. If motif dictionary has description amortized across instances, the effective bits per instance drop. Model as a divisor on $A(X)$.

Parallelism : independent sub-assemblies compress as a product code; wall-clock and effective complexity per unit time fall. Model as a divisor on $A(X)$.

Degeneracy : if many alternatives achieve the same function, the description of “a solution” is shorter than “this specific solution.” Function-first coding reduces effective complexity; model as a divisor.

This aligns your original formula with MDL-style accounting: EAIMR rises when, despite these discounts, the remaining coordinated specification is still large and functionally tight.

A.6 Coherence via Control & Causal Graphs

Let $G(X)$ be a causal graph extracted from X ’s dynamics. Define a control coherence term:

$$\phi(X) = \beta_1 \text{Stab}(G) + \beta_2 \text{GoalAcc}(G) + \beta_3 \text{Robust}(G)$$

GoalAcc(G): accuracy tracking a specified functional setpoint (e.g., temperature band in termite mounds).

Robust(G): performance under noise/lesions (graceful degradation).

Then use $\phi(X)$ inside (Section 3) or directly as your multiplier.

A.7 Practical Estimation (finite data)

Because $K(X)$ is uncomputable, rely on approximations:

Compression proxies: on suitable encodings (graphs, fields, symbol streams).

Surprisal under null: fit P_0 (e.g., generative physics sims, random graphs) and estimate by Monte Carlo or density models.

Function & coherence: quantify with control/evaluation tasks (tracking error, energy efficiency, resilience curves).

Discounts: measure from libraries (motif counts), concurrency (critical path vs. total ops), and function-equivalence classes (number of viable variants).

A.8 Worked Intuitions (triad examples)

Neural tissue: very high $A(X)$ (precise mesoscale organization), strong $C(X)$ (robust goal pursuit), discounts present (columns, parallelism), yet EAIMR stays high mind-like.

Termite mound: moderate $A(X)$ vs. dunes, high $C(X)$ (thermoregulation), sizable discounts (motifs, parallel labor, many workable designs) but coordinated control keeps EAIMR high.

Forest: extremely high macro $A(X)$ vs. random vegetation, high robustness/goal achievement (nutrient routing, succession), massive reuse/degeneracy, yet coherence over centuries pushes EAIMR high.

A.9 Decision Rule (operational)

Define thresholds from a reference set of clearly mindless assemblies:

$$\text{EAIMR}(X) \geq \tau \Rightarrow \text{classify as mind-like (candidate)}$$

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Takeaway

Information-theoretic view: EAIMR is a mind detector for improbably coherent order—high surprisal under a null, discounted for shortcuts, and boosted by functional integration.

It sidesteps narrativity bias: you don't need language or imagery to register as intelligent; termite colonies, forests, and atypical human cognition can score high if they assemble and maintain low-entropy, goal-directed structure against default dynamics.

B Derivations and Pseudocode

B.1 EAIMR Derivation

From surprisal: $S(X) = I_0(X) + \lambda\phi_{\mathcal{P}}(X)$, EAIMR as monotonic $g(S(X))$.

B.2 Pseudocode

```
def compute_eaimr(X, P0, P1, G_P):  
    A = assembly_index(X)  
    R = reuse_factor(X)  
    P = parallelism(X)  
    D = degeneracy(X)  
    C = coherence(G_P)  
    return (A / (R * P * D)) * C
```